Lab 5: From Linear Estimation to Learning Representations

Where We Are After DLQ4

So far, you have built models that look like this:

$$\hat{y} = ax_1 + bx_2 + cx_3 + d$$

This equation appeared in different disguises across labs:

- **Lab 1:** logical decision using weighted sum
- **Lab 2:** decision + confidence using sigmoid
- **Lab 3:** numeric prediction using a linear neuron
- **DLQ4:** age estimation using a single fixed equation

In every case:

- inputs were numeric
- output was produced by one equation
- learning meant changing the numbers, not the form

DLQ4 showed that this approach fails systematically for some problems.

Let's understand **why**, step by step.

1. What Exactly Is a "Model" So Far?

Up to now, a model has consisted of two parts:

1.1 Fixed Structure (Does Not Learn)

$$\hat{y} = f(x_1, x_2, \ldots, x_n)$$

This structure defines:

- number of inputs

- how they are combined
- whether output is linear or passed through a function

This structure **never changed** during learning.

1.2 Changeable Numbers (Learned)

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$$w_1, w_2, \ldots, w_n$$

are the **importance values** (weights)

$$b$$

is the **baseline offset** (bias)

Learning so far meant:

> Adjusting these numbers to reduce error.

2. What a Single Linear Equation Can Represent

General linear model:

$$\hat{y} = \sum_{i=1}^{n} w_i x_i + b$$

2.1 It Is Linear in the Inputs

If an input changes by a fixed amount:

$$\Delta \hat{y} = w_i \cdot \Delta x_i$$

This effect:

- does **not** depend on other inputs
- does **not** change across input ranges

That means:

- same slope everywhere
- same behavior for small and large values

This is called a **global linear relationship**.

2.2 Geometric Interpretation (Key Insight)

A linear model creates a **flat decision surface**:

- 1 input → straight line
- 2 inputs → plane
- 3 inputs → flat surface
- many inputs → **hyperplane**

A hyperplane can:

- shift
- rotate
- tilt

But it **cannot**:

- bend
- change slope in different regions
- represent “fast growth then slow growth”

That is exactly the failure observed in DLQ4.

3. Why Tuning Cannot Fix DLQ4

Students observed:

- fixing errors for young abalones worsened old ones
- fixing old ones worsened young ones
- similar weights produced very different ages

This is **not** because:

- learning rate was wrong

- coefficients were poorly tuned
- dataset was noisy

It happens because:

> One straight line is being forced to explain multiple behaviors.

No amount of tuning can make a straight line behave like a curve.

This is a **representation problem**, not a learning problem.

4. Measuring "How Wrong" a Prediction Is

For one data point:

$$\text{error} = y - \hat{y}$$

Where:

$$y$$

= true value

$$\hat{y}$$

= predicted value

But error alone is not enough.

4.1 Why the Sign Is Not Useful

Errors like:

$$+500 \quad \text{and} \quad -500$$

are equally bad.

So we measure **magnitude**, not direction.

5. Loss Functions (Quantifying Mistakes)

A loss function converts error into a **non-negative number**.

5.1 Mean Squared Error (MSE)

For one sample:

$$L = (y - \hat{y})^2$$

For N samples:

$$L = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

Why square?

- removes sign
- penalizes large mistakes more than small ones

This matches DLQ4 intuition:

> Small errors are acceptable. Large errors are problematic.

6. Learning as Error Reduction

Learning now means:

Change the numbers

$$w_i,\ b$$

so that the loss decreases.

But how do we know which direction to change them?

7. Gradient: Direction of Change

The gradient tells us:

> If I slightly change a parameter, does the loss increase or decrease?

For a weight:

$$\frac{\partial L}{\partial w_i}$$

If:

- gradient > 0 → increasing wiw_iwi increases loss
- gradient < 0 → increasing wiw_iwi decreases loss

So we move in the **opposite direction** of the gradient.

8. Gradient Descent Update Rule

For each weight:

$$w_i := w_i - \eta \frac{\partial L}{\partial w_i}$$

For the bias:

$$b := b - \eta \frac{\partial L}{\partial b}$$

Where:

$$\eta$$

is the **learning rate**, controlling step size.

This is not magic.

It is **controlled correction** based on slope.

9. Why Learning Still Fails in DLQ4

Even with:

- perfect loss definition
- correct gradients
- infinite training time

A single linear equation still produces:

- systematic bias
- repeated mistake patterns

Because:

> Gradient descent can only adjust numbers, not bend the surface.

10. The Real Limitation (Key Takeaway)

So far:

- inputs are combined once
- output is produced once
- learning only tunes coefficients

This means:

> Model complexity is fixed before learning begins.

To represent:

- different slopes in different regions
- growth that slows after maturity
- interactions between inputs

We need models whose **representation itself can change**.

11. Where This Is Heading

The next idea will be:

- applying multiple linear transformations
- introducing intermediate representations
- allowing the surface to bend

This does **not** mean:

- more tuning
- better loss

- different optimizer

It means:

> Changing the **shape** of the model, not just its position.

That is the motivation for:

- multilayer models
- backpropagation
- deep learning

Transition Line

> “Until now, learning meant changing numbers inside a fixed equation.
>
> Some problems fail because the equation itself is too simple.
>
> The next step is to let the model learn how to represent the problem.”